BrewLite | Deployment Diagram

ASSIGNMENT 2 - TARGET DESIGN | Vercel Hobby + Render + VNPay Sandbox | Live verification pending

PRODUCTION TARGET / PUBLIC HTTPS | Browser and VNPay are external

3 Browser -> signed VNPay payment URL (HTTPS)

<<device>>
Customer browser

UI from Vercel
JWT on protected API calls
No backend signing keys

1 / 4 HTTPS :443
UI + REST/JSON

<<executionEnvironment>> Vercel Hobby
Public frontend HTTPS :443

<<artifact>> Next.js prebuilt frontend
CI build -> Vercel assets / server runtime
/api/backend/* proxy + /payment

Vercel environment
NEXT_PUBLIC_API_URL (public)
BACKEND_INTERNAL_URL (server only)
Both target public Render HTTPS

4 Return -> browser
Then HTTPS GET /payment

<<external service>>
VNPay Sandbox

Hosted payment page
Server-to-server IPN
QueryDR API
Merchant settings: Return / IPN

6 QueryDR
Outbound HTTPS POST
Verified gateway result

5 IPN: server-to-server
HTTPS GET
/payments/vnpay/ipn

LOCAL / developer

Compose: web + API + DB
Isolated A2 DB; ignored .env
No production credentials

CD
HTTPS

2 REST / JSON over public HTTPS :443
Vercel proxy / SSR -> Render ingress

PRODUCTION / Render | region: TBD

Public HTTPS :443 / TLS ingress -> process $PORT
Render health: GET / -> 200 (liveness only)

<<CI environment>>
GitHub Actions runner
PLANNED - no workflow yet

PR: isolated DB + build/tests
CD: Vercel CLI build/upload
CD: trigger Render Docker build

GitHub Actions secrets (planned)
VERCEL_TOKEN + project IDs
RENDER_DEPLOY_HOOK_URL
No live DB/payment secrets

CD / HTTPS

<<executionEnvironment>> Render Web Service

<<artifact>> Docker runtime image (Render-built)
NestJS + Prisma Client; IPN + shared finalization
Order expiration + reconciliation timers

Render server environment - secrets
DATABASE_URL, JWT_SECRET, INTERNAL_SECRET
VNPAY_HASH_SECRET (values never shown)

PostgreSQL / TCP
:5432, private
Internal DB URL

RENDER PRIVATE NETWORK
PostgreSQL is not a public API

<<executionEnvironment>>
Render PostgreSQL Free

<<artifact>> schema + persisted data
Internal Database URL only
External access OFF (verify)
Same workspace / region as API

READ THIS AS A TARGET DESIGN

Environments: Local, isolated CI, Production target (not verified).
Solid arrows: target runtime communication (verify after deploy).
Purple dashed arrows: planned CI/CD deployment dependencies.
Green boundary: private DB path; no browser / Vercel DB link.
Secrets: Render server env; CI deployment keys in GitHub Secrets.
Return is navigation only. IPN / QueryDR verify before finalization.
Crossing lines without a junction do not indicate a connection.

BUILD / ARTIFACT FLOW (PLANNED)

Frontend: CI Vercel build -> prebuilt upload to Vercel.
Backend: Git SHA + backend/Dockerfile -> Render Docker build
-> Render-managed private image registry -> runtime.
No separate BrewLite image registry (GHCR / Docker Hub).
Task 5: tagged-image / release / rollback evidence required.
Live URLs / CI/CD / migration / payment E2E: still pending.